



Bharath Kumar Kagey

SR. MECHANICAL DESIGN ENGINEER
• HPDC TOOLING

NAVIGATION

- Portfolio
- Work Experience
- Education
- Resume ↓
- LinkedIn ↗
- GitHub ↗

CONTACT

- ✉ bharath.kagey@gmail.com
- 📍 Fremont, CA
- ☎ 313-283-1034

CORE TOOLS

- CATIA V5
- MAGMASOFT
- ANSYS
- 3DEXPERIENCE
- ENOVIA
- GD&T / RSS
- PowerShell
- RDnT
- NADCA

ENGINEERING PORTFOLIO

HPDC Tooling at GIGA Scale.

Senior Mechanical Design Engineer with 9+ years of OEM experience and 6 years leading HPDC die development at Tesla's GIGA Castings facility.

200k

DIE LIFE TARGET

43%

DIE LIFE IMPROVEMENT

67%

REFURB TIME REDUCTION

19%

CYCLE TIME IMPROVEMENT

ABOUT

About Me

I am a Senior Mechanical Design Engineer specializing in High Pressure Die Casting tooling, with a focus on die architecture design, thermal system optimization, and simulation-driven development at Tesla's GIGA Castings program. Over 6 years at Tesla I have led programs from concept through high-volume Model Y production — from ground-up die development targeting 200,000 shot life to PLM infrastructure rollout across two manufacturing sites. My work sits at the intersection of mechanical design, manufacturing systems, and engineering automation.

PROJECTS LED

Portfolio

DIE DEVELOPMENT • HPDC • TESLA GIGA CASTINGS

FRUB Die Development — 200,000 Shot Die Life Program



Bharath Kumar Kagey

SR. MECHANICAL DESIGN ENGINEER
· HPDC TOOLING

NAVIGATION

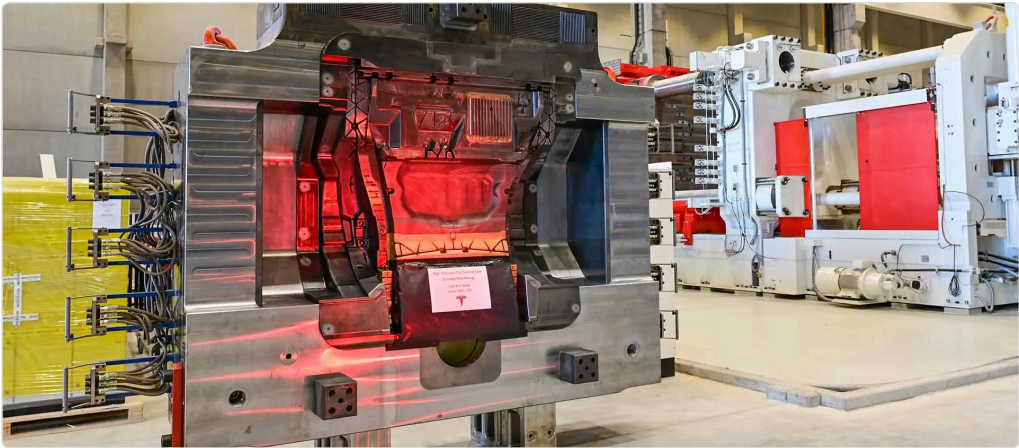
- Portfolio
- Work Experience
- Education
- Resume ↓
- LinkedIn ↗
- GitHub ↗

CONTACT

- ✉ bharath.kagey@gmail.com
- 📍 Fremont, CA
- ☎ 313-283-1034

CORE TOOLS

- CATIA V5
- MAGMASOFT
- ANSYS
- 3DEXPERIENCE
- ENOVIA
- GD&T / RSS
- PowerShell
- RDnT
- NADCA



Objective: Redesign the FRUB HPDC die from scratch to extend die life beyond the 140,000 shot predecessor, resolve chronic casting cracking in the motor cover and wheel house regions, and reduce cycle time.

Result: Achieved 200,000 shot die life (+43% over baseline), reduced cycle time from 105 sec to 85 sec (19% improvement), and eliminated chronic cracking defects through integrated architecture, cooling, and gating redesign.

140k → 200k shots (+43%)

105s → 85s cycle time

Refurb: weeks → days

Zone-by-zone steel coating spec

TECHNICAL APPROACH

- Introduced a dedicated insert holder with sub-inserted cavity blocks, replacing monolithic inserts with modular sections — each high-wear zone independently replaceable without full die strip, reducing unplanned downtime and extending overall die service life.
- Engineered conformal and 3D-printed cooling channels using BTU extraction calculations to achieve die face thermal balance, directly driving the 105s → 85s cycle time reduction at Model Y production rates.
- Redesigned runner and gating system to resolve melt fill deficiencies in motor cover and wheel house areas — controlled solidification front progression to eliminate crack initiation; validated in MAGMASOFT before cutting steel.
- Defined surface coating treatment and heat treatment parameters zone-by-zone with steel suppliers, targeting surface hardness improvement at high-wear gating and overflow zones — coordinated with BTU cooling layout to reduce thermal cycling stress.
- Integrated quick-connect 6061 aluminum cooling manifolds and brass couplers to eliminate stainless steel hard plumbing, cutting refurbishment turnaround from weeks to days.
- Cross-factory knowledge exchange with Giga Berlin engineering team — captured lessons learned from their FRUB die experience and incorporated proven improvement trials into the new design baseline.
- Full die assembly validated in MAGMASOFT (flow, thermal, solidification) and ANSYS (structural deformation, platen deflection, support pillar preload) prior to first steel cut.

MAGMASOFT

ANSYS Static Structural

CATIA V5

BTU Analysis

3D-Printed Cooling

6061 Al



Bharath Kumar Kagey

SR. MECHANICAL DESIGN ENGINEER
• HPDC TOOLING

NAVIGATION

- Portfolio
- Work Experience
- Education
- Resume ↓
- LinkedIn ↗
- GitHub ↗

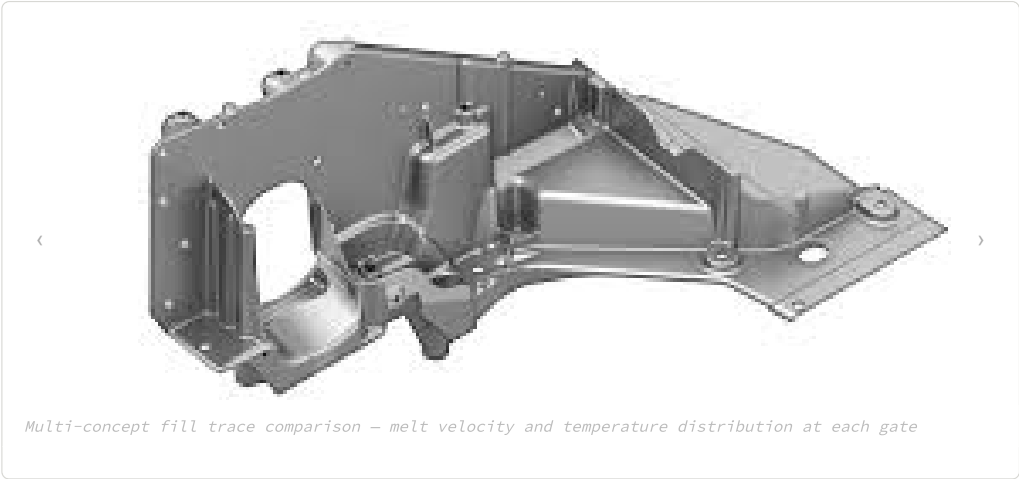
CONTACT

✉ bharath.kagey@gmail.com
📍 Fremont, CA
☎ 313-283-1034

CORE TOOLS

CATIA V5 MAGMASOFT ANSYS
3DEXPERIENCE ENOVIA
GD&T / RSS PowerShell RDnT
NADCA

Runner Design & Thermal Optimization — Torque Box, Rear Node & Shock Tower



Objective: Resolve chronic air entrapment on Torque Box, design a complete runner system for Rear Node dual-cavity die from scratch, and optimize the Shock Tower die thermal system to reduce cycle time and die soldering.

Result: Torque Box full shot weight reduced by 3.5 kg and cycle time from 85s to 68s. Rear Node launched through trim die buy-off into production. Shock Tower cycle time reduced from 100s to 90s with measurable reduction in die soldering and extended service life.

-3.5 kg full shot weight

85s → 68s (Torque Box)

100s → 90s (Shock Tower)

Rear Node launched to production

Reduced die soldering

TECHNICAL APPROACH

- **Gating & Flow:** Designed complete runner, gating, and venting for Rear Node dual-cavity die from scratch using NADCA and Buhler equations to correctly size gate velocity and vent area across both cavities; led through trim die buy-off. Ran multi-concept MAGMASOFT flow simulations on Torque Box redesign — evaluated melt velocity, temperature distribution, pressure trace, and fill pattern across candidate designs before any steel commitment.
- **Thermal Optimization:** Simulated die thermal systems for Torque Box, Rear Node, and Shock Tower in MAGMASOFT — mapped surface temperature distribution, identified hot spots driving die soldering and heat checking, and redesigned cooling circuit routing and sequencing to achieve balanced die face temperatures. Shock Tower cooling redesign achieved 100s → 90s cycle time reduction and extended insert service intervals.
- **Structural & Fixture:** Performed ANSYS structural analysis on cover/ejector halves and DCM platens to size support pillar preload, eliminating flash from die deflection under shot pressure. Designed a 6061 machined aluminum best-fit gauge fixture for Shock Tower castings enabling instant production-floor distortion measurement without CMM setup, accelerating root cause turnaround on quality holds.

MAGMASOFT

ANSYS

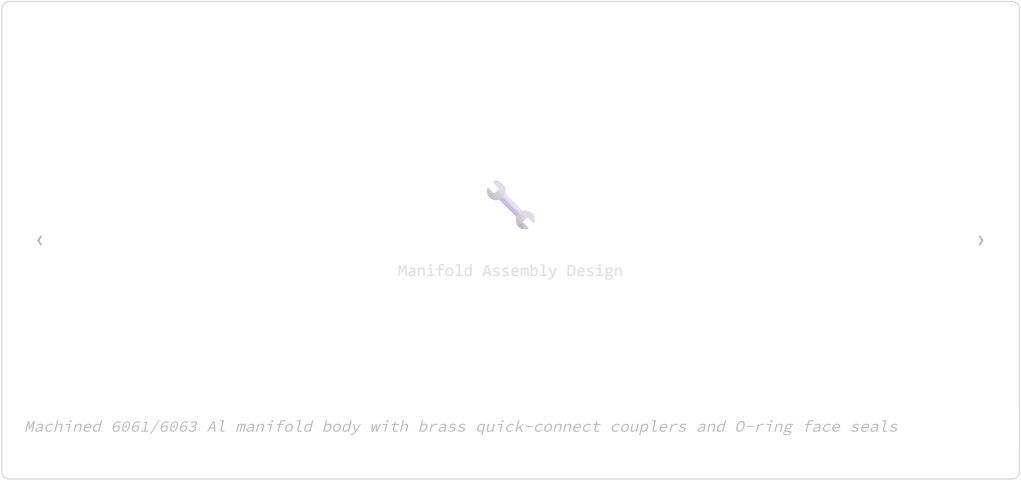
CATIA V5

NADCA Equations

Buhler Equations

6061 Al Fixture

Aluminum Manifold & Cooling Coupler Design — Die Refurbishment Optimization



Objective: Eliminate stainless steel hard plumbing with compression fittings — identified as the primary driver of 6-week die refurbishment cycles on GIGA dies.

Result: Die refurbishment turnaround reduced from 6 weeks to 2 weeks (67% reduction). Design subsequently adopted as the standard cooling interface on the FRUB die development program.

6 weeks → 2 weeks refurb

67% downtime reduction

Adopted on FRUB program

TECHNICAL APPROACH

- **Design:** Machined 6061/6063 Al manifold bodies with brass quick-connect couplers and O-ring face seals — replacing stainless steel hard plumbing with a standardized plug-and-play interface rated for HPDC cooling circuit pressures and temperatures. Design accounted for differential thermal expansion between the Al manifold and die steel body, swarf contamination ingress, and field-replaceable sealing elements.
- **End-to-End Ownership:** Produced all machining drawings, managed fabrication, led die installation, and trained technicians on connection procedure. Tried on Die S to validate sealing performance under production conditions before implementing on Die Q.
- **Impact & Adoption:** Reduced refurbishment turnaround from 6 weeks to 2 weeks by eliminating custom plumbing rework on every die swap. Technology adopted as the standard cooling interface on the FRUB die development program — demonstrating cross-program impact beyond original scope.

CATIA V5

6061/6063 Al

Brass Couplers

O-Ring Seals

GD&T Standards Development — Tesla GIGA Die Families



Bharath Kumar Kagey
SR. MECHANICAL DESIGN ENGINEER
• HPDC TOOLING

NAVIGATION

- Portfolio
- Work Experience
- Education
- Resume ↓
- LinkedIn ↗
- GitHub ↗

CONTACT

✉ bharath.kagey@gmail.com
📍 Fremont, CA
☎ 313-283-1034

CORE TOOLS

CATIA V5 MAGMASOFT ANSYS
3DEXPERIENCE ENOVIA
GD&T / RSS PowerShell RDnT
NADCA



Bharath Kumar Kagey
SR. MECHANICAL DESIGN ENGINEER
• HPDC TOOLING

NAVIGATION

- Portfolio
- Work Experience
- Education
- Resume ↕
- LinkedIn ↗
- GitHub ↗

CONTACT

- ✉ bharath.kagey@gmail.com
- 📍 Fremont, CA
- ☎ 313-283-1034

CORE TOOLS

- CATIA V5
- MAGMASOFT
- ANSYS
- 3DEXPERIENCE
- ENOVIA
- GD&T / RSS
- PowerShell
- RDnT
- NADCA



Datum Scheme Framework

Three-plane datum structure for GIGA die inserts – balancing thermal expansion, flash, and replaceability

- Objective:** Establish a unified GD&T and datum scheme standard for Tesla GIGA die tooling, balancing three competing demands: in-production thermal expansion, flash prevention at parting surfaces, and insert interchangeability during refurbishment.
- Result:** Standard adopted as governing reference in ENOVIA/PDM for all GIGA die releases. Saved 2 hours per insert replacement in the tool shop and reduced supplier drawing interpretation errors during die buy-off.

- 2 hrs saved per insert swap
- RSS statistical method
- Governing ENOVIA reference

TECHNICAL APPROACH

- **Standard Scope:** Authored GD&T and datum scheme standard balancing three competing functional demands simultaneously — thermal expansion compensation across HPDC cycling temperatures (~200°C delta), flash prevention at parting surface interfaces under 6200-ton clamp force, and insert-to-holder fit tolerances enabling drop-in replacement without hand fitting or shimming.
- **Tolerance Method:** Applied RSS (Root Sum Square) statistical tolerance analysis to determine probable assembly variation across the die insert stack, avoiding unnecessarily tight tolerances that inflate supplier machining cost with no quality benefit over worst-case arithmetic stackup. Output directly sized locating features and interface clearances.
- **Deployment:** Integrated standard into ENOVIA/PDM as the governing reference for all GIGA die releases across Lathrop and Fremont. Consistent supplier drawing interpretation reduced revision cycles at die buy-off; defined insert datum schemes saved 2 hrs per replacement in the tool shop by eliminating fit-and-adjust.

- GD&T
- RSS Tolerance Analysis
- ENOVIA/PDM
- CATIA V5

ENGINEERING AUTOMATION • POWERSHELL • CATIA API • AI-ASSISTED

Automated Sub-Insert Drawing Generation — PowerShell / CATIA Pipeline



Bharath Kumar Kagey

SR. MECHANICAL DESIGN ENGINEER
• HPDC TOOLING

NAVIGATION

- Portfolio
- Work Experience
- Education
- Resume ↓
- LinkedIn ↗
- GitHub ↗

CONTACT

- ✉ bharath.kagey@gmail.com
- 📍 Fremont, CA
- ☎ 313-283-1034

CORE TOOLS

- CATIA V5
- MAGMASOFT
- ANSYS
- 3DEXPERIENCE
- ENOVIA
- GD&T / RSS
- PowerShell
- RDnT
- NADCA

Bharath Kumar Kagey | Engineering Portfolio



Pipeline Architecture

3D CAD model → PowerShell script → auto-populated 2D drawing with GD&T annotations

Objective: Eliminate manual drafting of 20–50 sub-insert drawings per die set — a process taking up to 2 hours per drawing — by building an automated 3D-to-2D drawing generation pipeline.

Result: Per-drawing turnaround reduced from up to 2 hours to near-instant output. Pipeline deployed to supplier drawing releases as production-quality deliverables, validating output against Tesla's production drawing standard.

- 20–50 drawings per die set
- Up to 2 hrs → instant
- Deployed to supplier releases

TECHNICAL APPROACH

- **Pipeline Architecture:** Built a 3D-to-2D drawing automation pipeline using PowerShell scripting against the CATIA V5 COM API, with AI-assisted code generation (Claude) to develop and iterate scripts against specific engineering requirements. The pipeline reads sub-insert CAD geometry, extracts view projections, and auto-populates drawing templates with dimensional and annotation data.
- **Engineering Requirements:** Defined all automation logic against the internally developed GIGA die GD&T standard — datum scheme placement, feature control frame callouts, tolerance values by insert zone, title block field population, and revision table logic. Output covers sub-insert production drawings and ignition schematics: 20–50 drawings per die set.
- **Validation & Deployment:** Deployed pipeline output directly to supplier drawing releases — output met Tesla's production drawing quality bar, validating this as an engineering deliverable pipeline rather than an internal productivity tool. Per-drawing turnaround reduced from up to 2 hours to near-instant generation.

- PowerShell
- CATIA V5 API
- AI-Assisted Scripting
- GD&T Automation

PLM IMPLEMENTATION • 3DEXPERIENCE • LATHROP & FREMONT

3DEXPERIENCE PLM Implementation — GIGA Castings Tooling



PLM Migration Architecture



Bharath Kumar Kagey
SR. MECHANICAL DESIGN ENGINEER
• HPDC TOOLING

NAVIGATION

- Portfolio
- Work Experience
- Education
- Resume ↓
- LinkedIn ↗
- GitHub ↗

CONTACT

- ✉ bharath.kagey@gmail.com
- 📍 Fremont, CA
- ☎ 313-283-1034

CORE TOOLS

- CATIA V5
- MAGMASOFT
- ANSYS
- 3DEXPERIENCE
- ENOVIA
- GD&T / RSS
- PowerShell
- RDnT
- NADCA

OneDrive unstructured file storage → 3DEXPERIENCE structured PDM across Lathrop and Fremont

Objective: Replace unstructured OneDrive file sharing with revision-controlled PDM for the GIGA Castings Tooling team across Lathrop and Fremont — establishing the first structured data management system for the tooling organization.

Result: Full migration completed across both sites. CAD standards, standard component library, and auditable release workflow deployed. Dassault engaged directly to resolve large-dataset migration challenges.

- 2 sites migrated
- OneDrive → structured PDM
- First tooling PDM at Lathrop

TECHNICAL APPROACH

- **Migration:** Led full migration of GIGA Castings tooling CAD from unstructured OneDrive storage to 3DEXPERIENCE PLM across Lathrop and Fremont — establishing revision-controlled, searchable data management for the first time on the tooling team. Worked directly with Dassault to resolve large-dataset migration challenges and ensure cross-site data integrity.
- **Standards & Library:** Authored formal CAD standards documentation covering structure, naming conventions, and release procedures; configured a standard component library within 3DX to standardize common die components and tolerances across GIGA die families, reducing redundant design work program-to-program.
- **Workflow:** Created a dedicated Castings Tooling collaborative space in 3DX with a customized design release workflow suited to tooling-specific approval requirements — replacing ad hoc file sharing with a structured, auditable release process traceable to die revision history.

- 3DEXPERIENCE
- ENOVIA/PDM
- CATIA V5
- Dassault Systèmes

WORK EXPERIENCE

Work Experience

JULY 2020 – PRESENT

Tesla Motors — Lathrop, CA

Sr. Tooling Engineer – HPDC | Acting Engineering Supervisor (Jan 2023 – Jul 2025)

- Designed runner, gating, and venting systems for HPDC dies including Torque Box and Rear Node dual-cavity programs using NADCA and Buhler equations — reducing Torque Box shot weight by 3.5 kg and cycle time from 85s to 68s.
- Performed MAGMASOFT flow and thermal simulations for Lathrop Model S/X dies, optimizing runner geometry, cooling layout, and thermal balance.
- Introduced die longevity technologies: 3D-printed conformal cooling inserts, advanced steel alloys, controlled coating and heat treatment specs.
- Managed 5 Tool & Die technicians across two 6200-ton DCMs; led production launches for RUB, Torque Box, Shock Tower, and Rear Node GIGA dies.
- Designed 6061 Al manifold bodies with brass quick-connect couplers replacing hard plumbing — cutting refurbishment from 6 weeks to 2 weeks.

JULY 2018 – MAY 2020



Bharath Kumar Kagey

SR. MECHANICAL DESIGN ENGINEER
• HPDC TOOLING

NAVIGATION

Portfolio

Work Experience

Education

Resume ↓

LinkedIn ↗

GitHub ↗

CONTACT

✉ bharath.kagey@gmail.com

📍 Fremont, CA

☎ 313-283-1034

CORE TOOLS

CATIA V5 MAGMASOFT ANSYS

3DEXPERIENCE ENOVIA

GD&T / RSS PowerShell RDnT

NADCA

Segula Technologies (Volvo Trucks NA) — Greensboro, NC

Design Engineer – CAB | Geometrical Assurance Engineer | Fixture Design Engineer

- ▶ Maintained CAB design for Volvo and Mack Trucks across early and late-stage releases; managed ECNs for structural and design releases in CATIA V5 and Windchill PLM.
- ▶ Led front steel bumper design for Mack Anthem from concept through production — brackets, plastic deflectors, and all installation drawings.
- ▶ Performed GAE on New VNL chassis fairing brackets using RDnT — datum schemes, RSS tolerance stack-ups, and 3D/2D design recommendations.
- ▶ Designed production fixtures for Mack Macungie Plant assembly lines including engine rack, turbo pipe, and floor mat drilling fixtures.

○ FEB 2014 – DEC 2015

Primark Infotech — India

Product Designer

- ▶ Designed plastic injection molded components for automotive CAB interior trim in CATIA V5 — draft analysis, parting line, clash analysis, DFM/DFA validation.
- ▶ Generated GD&T drawings, managed BOM, performed DRBFM failure mode analysis across multiple interior trim subsystems.

EDUCATION

Education



Master of Science, Automotive Engineering

Lawrence Technological University

December 2017



Bachelor of Science, Automotive Engineering

Osmania University

May 2015